

Module 4 – Authentication & Deployment

1. JWT Authentication

What is JWT?

JWT (JSON Web Token) is a secure way to transmit user authentication data between client and server.

Why JWT?

- Stateless authentication
- Secure data transfer
- Easy to scale
- Used in modern web apps

How JWT Works

- User logs in
- Server creates token
- Token sent to client
- Client uses token for requests

Example:

```
const jwt = require("jsonwebtoken");  
  
const token = jwt.sign({ id: 1 }, "secretKey", { expiresIn: "1h" });
```

2. User Login System

What is Login System?

A login system verifies user identity using email/username and password.

Steps

- User enters credentials
- Server checks database
- Password is verified
- JWT token is generated

Example:

```
app.post("/login", (req, res) => {  
  const { email, password } = req.body;  
  res.send("User Logged In");  
});
```

3. Security Best Practices

Why Security Matters

Security protects applications from unauthorized access and attacks.

Best Practices

- Hash passwords (bcrypt)
- Use HTTPS
- Validate input data
- Protect routes with JWT
- Avoid exposing sensitive data

Example:

```
const bcrypt = require("bcrypt");  
  
const hashedPassword = bcrypt.hashSync("mypassword", 10);
```

4. Deployment Basics

What is Deployment?

Deployment means making your application live on the internet.

Common Platforms

- Vercel
- Netlify
- Heroku
- Render

Steps

- Build project
 - Upload to server
 - Configure environment variables
 - Start application
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5. Cloud Deployment

What is Cloud Deployment?

Cloud deployment means hosting applications on cloud servers.

Popular Cloud Services

- AWS
- Google Cloud
- Microsoft Azure

Benefits

- Scalable
 - Reliable
 - Accessible worldwide
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6. Key Concepts to Remember

- JWT handles authentication securely
 - Login system verifies users
 - Security protects applications
 - Deployment makes apps live
 - Cloud hosting ensures scalability
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7. Summary

Authentication and deployment are essential for real-world backend applications. JWT ensures secure login systems, security practices protect data, and deployment brings applications to production-ready environments.